

Master Degree in Machine Learning for Health
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Master Thesis

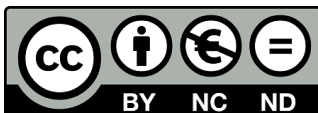
“Deep Learning for Photomultiplier Crystal Detection Signals”

Miguel Escudero Rodríguez

David Izquierdo-García
Leganés, September 2026

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Deep Learning for Photomultiplier Crystal Detection Signals

Miguel Escudero Rodríguez

Master's Degree in Machine Learning for Health

Universidad Carlos III de Madrid, Leganés, Spain

Email: 100569281@alumnos.uc3m.es

Abstract—**TODO:** 150–250 palabras. Problema, metodo propuesto, datos empleados y resultados cuantitativos principales. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

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Index Terms—Positron emission tomography, silicon photomultiplier, missing data imputation, deep learning, graph neural networks, silicon photomultiplier failure.

1 INTRODUCTION

Positron Emission Tomography (PET) is a non-invasive molecular imaging modality and one of the central techniques of nuclear medicine, providing an *in vivo* assessment of the metabolic and biochemical state of a patient [1]. Unlike purely anatomical modalities, PET relies on radiotracers: biologically active molecules chemically labelled with positron-emitting radionuclides [2]. Once administered, the tracer distributes according to the physiological pathway targeted by its carrier molecule, so that the measured signal reflects function rather than structure. When an emitted positron annihilates with an electron in the surrounding tissue, two photons of 511 keV — the rest mass energy of the electron — are released in nearly opposite directions. Detecting both of them in coincidence defines a line of response (LOR) along which the annihilation

took place, and the accumulation of millions of LORs allows the three-dimensional distribution of activity to be reconstructed [1]. This detection scheme gives PET a functional sensitivity that few other non-invasive imaging techniques can match [3], at the price of the tracer itself: positron-emitting radionuclides are radioactive by nature, costly to produce, and short-lived, which ties any PET facility to a nearby cyclotron and specific facilities [2].

This functional readout is most valuable where disease alters metabolism before it alters anatomy. In oncology, ^{18}F -FDG PET has become a cornerstone for diagnosis, staging, restaging and the assessment of treatment response [4], [5]. Its role in neurology is arguably even more distinctive, since it reveals pathologies that produce no macroscopic structural abnormality on conventional Magnetic Resonance Imaging (MRI) or Computed Tomography (CT), such as the characteristic hypometabolic and amyloid patterns of Alzheimer's disease [6], among other neurodegenerative and cerebrovascular disorders.

The sensitivity of PET is not matched by its spatial resolution. Clinical scanners typically resolve 4–5 mm full width at half maximum (FWHM) [7], and recent advances push the number towards 3 mm [7], [8], higher than the numbers normally achieved with MRI or CT. This limit decomposes into four contributions, two physical and two belonging to the detector, which behave very differently from one another [9]. The physical ones are the finite distance the positron travels before annihilating, set by the emitting isotope [10], and the small angular deviation from antiparallel emission, caused by the residual momentum of the electron–positron pair. Neither can be engineered away, although the second translates into a positional error that scales with the diameter of the detector ring and is therefore smaller in compact systems. The detector terms are more tractable. In pixelated arrays, the size of the individual element fixes the finest scale at which an interaction can be localised; shrinking it improves resolution, but the inter-crystal reflectors then occupy a growing fraction of the detector volume, and the probability that a photon deposits energy across several elements rises. The decoding error — the uncertainty in determining where within the module the interaction occurred — amounts to roughly one third of the detector width [9]. Unlike positron range and non-collinearity, this term is not a physical floor but a consequence of how the light distribution is read and interpreted, and it is therefore the one most open to improvement, whether through detector geometry, readout

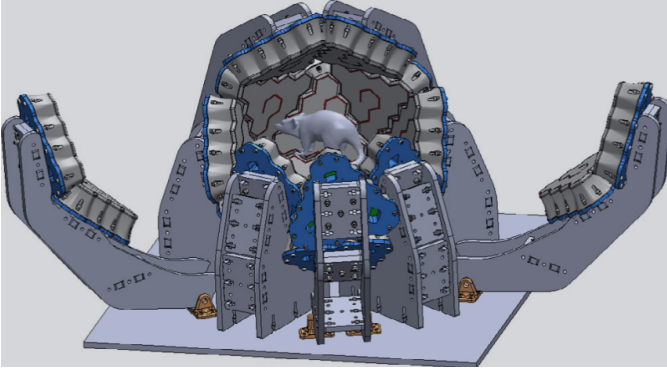


Figure 1: **TEMPORAL** Three-dimensional model of the quasi-spherical, hexagonal-based PET scanner. Reproduced from [12].

electronics or the algorithms that interpret the signal. This is the ground occupied by organ-dedicated scanners. Adapting the detector geometry to a specific anatomical region increases solid-angle coverage, and with it sensitivity and resolution, at a lower manufacturing cost than a general-purpose whole-body system; designs exist for breast, brain, prostate and cardiac imaging [11]. The trade-off is that irregular geometries complicate calibration and data correction, and that small gantry apertures accentuate parallax errors and degrade the uniformity of spatial resolution across the field of view [11]. Much of the design effort in these systems therefore shifts onto the detector itself.

Our group is currently developing a quasi-spherical brain-dedicated PET scanner along these lines [12]. Rather than the cylindrical ring of conventional systems, its detector modules tile a closed surface around the head (Fig. 1), approaching full 4π steradian coverage and thereby collecting a far larger fraction of the emitted coincidences [12]. The modules are hexagonal, a shape chosen for two independent reasons: hexagons tile a surface without gaps and approximate a sphere more closely than square modules of comparable size, and hexagonal scintillators have also been shown to offer slightly better light yield and energy resolution than square, triangular or cylindrical prisms [13]. Each module is built around a monolithic LYSO crystal [14].

The choice between segmenting the scintillator and leaving it continuous is not incidental, and it conditions how the interaction position can be recovered at all [15]. In pixelated arrays, the crystal is cut into individual elements separated by reflectors, so that the interaction is assigned to whichever element fired. Event positioning is then simple and fast, but the resolution cannot be finer than the element itself [9], and two costs follow from making the elements smaller: the reflectors occupy a growing fraction of the detector volume, reducing the packing fraction, and the number of readout channels rises. Multiplexing schemes, in which several crystals share a readout line and the position is recovered by decoding the light distribution, are commonly used to keep this cost manageable [11]. Pixelated arrays also provide no intrinsic

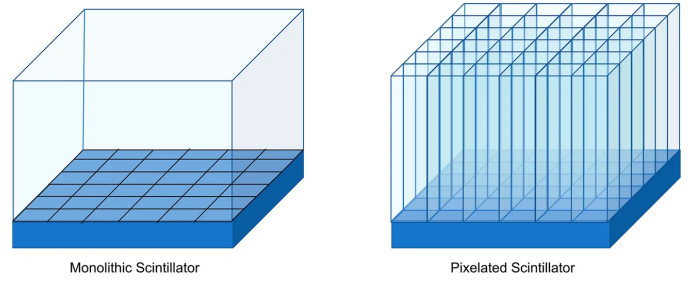


Figure 2: Monolithic (left) versus pixelated (right) scintillator configurations. In the monolithic design a continuous slab is coupled to a segmented photosensor array; in the pixelated design the crystal is cut into elements coupled one-to-one to the photosensors. Reproduced from [17] under CC BY 4.0.

depth-of-interaction (DOI) information, which has motivated layered designs such as phoswich detectors, where stacked scintillators with different decay times allow a coarse depth assignment at the price of a degraded timing resolution [11].

Monolithic scintillators take the opposite approach. A single continuous block is coupled to an array of photodetectors, and the light from each interaction spreads over many channels at once [15]. Positioning is no longer a matter of identifying an element but of estimating a coordinate from the recorded light distribution, which shifts the resolution limit from the geometry of the crystal to the quality of the positioning algorithm, and makes the DOI recoverable as a continuous variable [16]. The absence of internal reflectors also restores the packing fraction lost in segmented designs [11]. This is the configuration adopted in the scanner described above, and it is the light sharing across the array that the present work exploits.

This light distribution is read out by an array of silicon photomultipliers (SiPMs) optically coupled to the rear face of the crystal [15]. Each interaction produces signal in a subset of channels, and a readout application-specific integrated circuit (ASIC) amplifies, thresholds and digitises each of them, so that an event is fully described by the vector of charges recorded across the array. Because a channel only produces an output once it exceeds the threshold set in the electronics, most entries of that vector are zero for any given event and only a few channels contribute; the sparsity of the signal is therefore a property of the acquisition system rather than a defect of the data [18].

Estimating the interaction position from that vector admits solutions of widely varying complexity. The most common is the centre of gravity (CoG), which places the event at the average of the photodetector coordinates, each weighted by the charge that channel recorded [19]; in practice the square of the charge is normally used as the weight, which reinforces the contribution of the channels closest to the interaction point and improves resolution [20]. Its appeal is that it is deterministic, has no parameters to tune and requires no prior calibration. In exchange, it exploits only part of the information contained in the light distribution: recovering the DOI, or correcting the

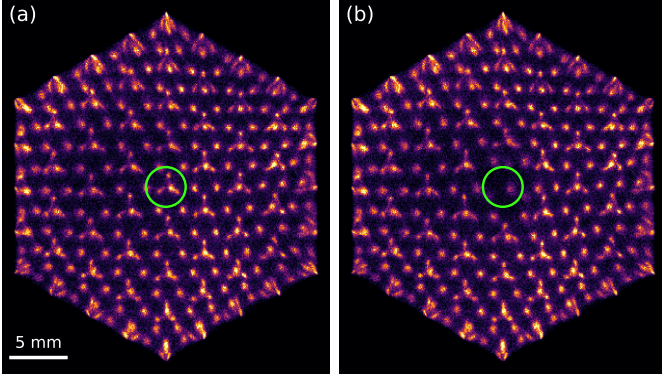


Figure 3: Flood map of the same detector module reconstructed by centre of gravity, using the same events in both panels: (a) with all 61 channels operational and (b) with one channel disabled, marked in green. The structures engraved in the crystal appear as discrete spots; the inset magnifies the neighbourhood of the disabled channel.

bias the method itself introduces near the crystal edges, calls for substantially more elaborate algorithms [21].

The two-dimensional histogram of the positions computed in this way over a large number of events forms the flood map (Fig. 3), in which the structures engraved in the crystal appear resolved as discrete spots. It is the image conventionally used to characterise a detector of this kind, and its sharpness depends on all 61 channels being operational: the CoG averages over the whole array, and a single missing term is enough to displace the centroid. The effect is confined to the neighbourhood of the failing channel — in the module of Fig. 3 the reconstructed position shifts by a median of 0.41 mm for events within 4 mm of it, and by less than 0.01 mm elsewhere — but within that neighbourhood the engraved structures are no longer resolved.

Silicon photomultipliers degrade and fail for a variety of reasons: thermal drift, manufacturing defects, deterioration of the optical coupling, or faults in the readout channel. The consequence for the image is well recognised in industrial practice: a dead pixel introduces reconstruction artefacts and incorrect activity values, and commercial systems include quality-control procedures to identify it [22]. The adopted general solution, however, is to discard the affected information and compensate for it downstream of positioning, either by duplicating events from neighbouring pixels in list mode or by interpolating in sinogram space [22]. The channel is not recovered: it is masked out.

This class of problem has motivated the application of deep learning to PET detector readout [23]. The most developed line is the estimation of the interaction position in monolithic crystals, where networks trained on calibration data consistently outperform analytical algorithms and recover the DOI as a continuous variable [24], [25]. Two previous studies address precisely this task on the present detector [18], [26]. All of them share the same framing: the light distribution recorded by

the array is taken as input and the network returns a coordinate.

Closer to what is proposed here, deep learning has also been applied to recovering channel signals rather than positions. Networks have been trained to invert multiplexed readout schemes, reconstructing the individual channel values from a reduced set of measurements and validating the result on the recovered flood map [27], [28], and the approach has recently been extended to semi-monolithic detectors, where the full 64-channel response is reconstructed from 16 row-and-column sums by exploiting the light sharing between slabs [29]. In all these cases, however, the information carried by each channel is still present in the acquired data: multiplexing encodes it in a known linear combination, and the network learns to invert a mapping fixed by design. A failed channel is a different situation, since nothing of its signal reaches the acquisition and the only remaining trace of it is the statistical correlation that light sharing induces among its neighbours.

That situation is not unique to PET. Recovering the response of defective elements in a sensor array is an established problem in imaging, addressed either by interpolating from neighbouring pixels or, more recently, by training autoencoders to reconstruct corrupted regions from their surroundings [30]. The formulation adopted here belongs to that family: it is the masked autoencoder setting [31], in which part of the input is deliberately hidden and the model is trained to reconstruct it, which allows supervision to be generated from healthy modules without requiring labelled faulty detectors. What distinguishes the present case is that the redundancy being exploited is physical rather than statistical: the light emitted in a single interaction is measured simultaneously by several photodetectors, so the value of a channel is constrained by the response of those around it.

This work proposes to recover that channel rather than discard it. If the light from each interaction spreads over several photodetectors, the charge that the failed sensor would have recorded is not entirely lost: it remains implicit in what its neighbours measured, and the problem is to estimate it. The difficulty lies less in the model than in the supervision. In a faulty module there is no way of checking whether an estimate is correct, since the true charge of the dead channel is unobservable by definition: if it could be measured, the channel would not be faulty. What cannot be observed, however, can be simulated. It suffices to take a healthy module, disable one of its channels and keep the original value as a reference; the fault is fabricated, and with it the label. The whole approach rests on that idea, and it is applied to experimental data acquired with the detector described above rather than to simulations, so that the model learns the behaviour of the system as it actually is, with its noise, its threshold and its variation between modules.

The choice of model follows from the geometry of the array. The immediate option would be to treat the vector of sixty-one charges as a list of numbers and let a dense network discover on its own which channels are related to which, but that means learning from scratch something that is already known: which sensor is adjacent to which. The photodetectors form a hexagonal tiling in which every interior sensor has six

immediate neighbours, and that neighbourhood is precisely the path along which the light is shared. A structure of this kind is naturally described as a graph, with one node per sensor and an edge between each pair of adjacent sensors, and it therefore admits a neural network that operates on it by propagating information between neighbours [32]. The model proposed here follows that approach, incorporating the geometry of the detector as prior knowledge instead of leaving it implicit in the data.

The evaluation is framed along the same lines. The error in the estimated charge is the quantity the model minimises, but on its own it says little about the usefulness of the method: what matters is how much is recovered of what the fault degrades. Three physical effects are therefore measured: the displacement of the reconstructed position, which is the direct consequence on the flood map; the conservation of the energy spectrum, which determines whether the restored signal is physically admissible; and the spatial resolution. All of this on modules the network has not seen during training, and compared against classical interpolation methods in order to establish that the complexity of the model is warranted. The behaviour of the method when more than one channel fails is also examined, with particular attention to contiguous groups, which are what is observed in real faulty detectors and also the least favourable case, since when an entire region disappears the lost sensor is left without healthy neighbours to draw on.

2 MATERIALS AND METHODS

2.1 Detector and Data

Every measurement reported here comes from detector modules of the scanner described in Section 1. Each module is built around a monolithic LYSO hexagonal prism, 17.6 mm on a side and 13 mm thick, laser-etched into 364 optical structures that constrain how the scintillation light spreads before reaching the photosensor [14]. An array of silicon photomultipliers is optically coupled to its posterior face [12]. The array provides 61 instrumented readout channels, which tile a hexagonal lattice with a pitch of 3.75 mm and span 26 mm across, so that the sensors do not reach the crystal boundary. The lateral faces of the crystal carry an absorbing black adhesive that suppresses edge reflections. This has a consequence that recurs throughout our results: interactions near the perimeter lose a measurable fraction of their light, which shifts both the charge recorded by the peripheral channels and the position of the energy photopeak in those regions.

We used acquisitions recorded with a ^{22}Na source under flood irradiation, one file per physical module, comprising 159 healthy modules and 62 modules with at least one suspected faulty channel. Each file holds on the order of 10^6 events. An event activates 17 channels on average, ranging from a single channel to 54, since a channel only produces an output once it exceeds the threshold set in the readout electronics. We applied neither energy calibration nor an energy window. Both would require a per-channel calibration that is not available for this set of modules, and the metric we use to assess spectral fidelity

is constructed so that their absence cancels, as Section 2.6 explains.

We split the healthy modules by file rather than by event, with a fixed seed: 149 for training, 5 for validation and 5 held out for testing. The distinction matters more than it might appear. Each file is a different physical detector, with its own gain, its own optical coupling and its own defects, so splitting by file measures generalisation to hardware the model has never seen. Splitting by event would let the model learn the idiosyncrasies of a detector and then be tested on that same detector, which is a considerably easier problem and not the one that arises in practice. The test partition was reserved before any training and used neither for fitting nor for model selection.

2.2 Preprocessing and Sample Generation

The raw quantity is a vector $\mathbf{r} \in \mathbb{R}^{61}$ holding the digitised charge of each channel for one event. We applied two operations before it reaches the network. Occasional negative values, produced by baseline subtraction in the electronics, were clipped to zero, since a negative charge has no physical meaning and would distort the centroid. Events in which no channel exceeded the threshold carry no information and were discarded.

We then simulated a faulty channel by forcing a subset $\mathcal{D}$ of the entries to zero and asking the model to recover the original vector from what remained. The network additionally receives a binary mask indicating which entries were suppressed, so that a channel switched off is distinguishable from a channel that legitimately recorded no signal. This is the masked autoencoder setting [31] applied to detector channels instead of image patches, and it makes the supervision self-generated: the target is the original vector, so no module with a genuinely dead sensor is needed for training. We drew a fresh $\mathcal{D}$ for every sample, so a given event is never seen twice with the same failure pattern.

Normalisation, however, deserves a comment, because it constrains the architecture. We scaled each event by the maximum of its own masked vector,

$$\mathbf{x} = \frac{\tilde{\mathbf{r}}}{\max_c \tilde{r}_c}, \quad \mathbf{y} = \frac{\mathbf{r}}{\max_c r_c}, \quad (1)$$

where $\tilde{\mathbf{r}}$ is the masked vector and $\mathbf{y}$ the target. Two choices are embedded here. Scaling per event rather than per dataset follows from the physics: for position estimation what matters is how the light distributes across the array, not how much light the interaction released. Computing the factor *after* masking means that the target of a suppressed channel exceeds unity whenever that channel was the brightest of the event. The output layer is therefore linear, with no sigmoid or clamp, because bounding it would make the brightest failures unrecoverable by construction.

One further choice concerns which channel to suppress. Roughly three quarters of the channels in any given event are below threshold, so drawing the seed uniformly would produce a training set in which most samples require no correction at

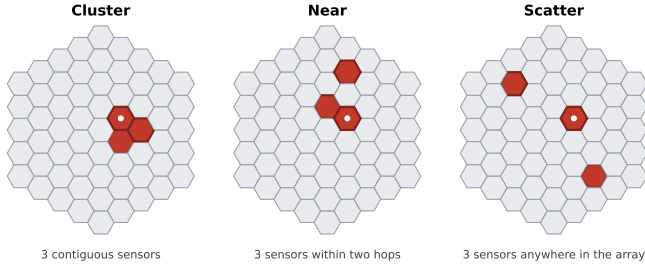


Figure 4: The three failure regimes used to generate the training masks, shown on the 61-sensor array for $k = 3$. Suppressed sensors are marked in red and the white dot indicates the seed from which the pattern grows. Cluster forces the failures to be contiguous, scatter distributes them across the array, and near places them within two hops of the seed without requiring contact.

all, and the model would reasonably learn to predict zero. We therefore balanced the draw: half of the samples suppress a channel that carried signal and half suppress a channel already at zero.

2.3 Failure Regimes

A single dead channel is the simplest fault, but not the only one. When we inspected the flood maps of the 62 defective modules, we found three recurring patterns. What matters is not so much how many sensors fail as how they are arranged: one whose neighbours are all healthy can be reconstructed from them, whereas one surrounded by other dead sensors cannot.

We reproduced those patterns with three mask-generation strategies, illustrated in Fig. 4. In the *cluster* regime the k suppressed channels grow outward from the seed so that they remain contiguous, which is the least favourable case: an entire region of the array disappears and the affected sensors are left without healthy neighbours to draw on. In the *scatter* regime they are drawn uniformly across the array, so each failure is effectively isolated. The *near* regime sits between the two: the k channels lie within two hops of the seed without being required to touch. That intermediate case is the one we observed most often in real faulty modules, typically two or three sensors separated by one or two positions, and it is the reason we introduced it rather than settling for the two extremes.

2.4 Network Architecture

The geometry of the array suggests how to model the problem. A dense network receiving the 61 charges as a flat vector would have to learn which sensors are adjacent, something already known from the physical layout. We therefore represented the detector as a graph whose nodes are the 61 active sensors and whose edges join physically adjacent ones, and propagated information along those edges [32], [33]. The resulting adjacency has 312 directed edges, with 37 nodes of

degree six, 18 of degree four and 6 of degree three, and it is fully symmetric.

Each layer of our model, which we refer to as HexGNN, updates a node from its own state and the mean of its neighbours,

$$h_c^{(l+1)} = \sigma \left(W_s h_c^{(l)} + W_n \frac{1}{|\mathcal{N}(c)|} \sum_{j \in \mathcal{N}(c)} h_j^{(l)} + b \right), \quad (2)$$

where $\mathcal{N}(c)$ are the neighbours of sensor c and σ is a Gaussian error linear unit (GELU) [34]. The aggregation is isotropic: every neighbour of a node shares the weight matrix W_n , so the layer encodes adjacency but not direction. Batch normalisation [35] is applied between blocks. Our reference configuration uses a hidden width of 48 and four blocks, for 38 305 trainable parameters.

We compared this model against three groups of alternatives. As classical baselines without learned parameters we used the mean of the neighbouring channels and a per-channel linear regression on the remaining 60, both fitted on training modules only and both excluding the target channel from their inputs. As neural baselines without a geometric prior we trained a deep multilayer perceptron and a residual multilayer perceptron with channel attention. Both receive the same information but no notion of adjacency, which isolates the contribution of the prior from that of raw capacity.

The aggregation in Eq. 2 is deliberately simple, and it is fair to ask whether that simplicity is what limits performance. We therefore tested four alternatives, each addressing a different suspicion about where the bottleneck might be.

If the six neighbours of a sensor contribute unequally — and they might, since the light reaching one of them depends on the direction of the interaction — then averaging them discards useful information. Graph attention networks (GAT) [36] learn a weight per neighbour instead of fixing it, which lets the model decide how much each one should count. A related but broader concern is that the mean is only one summary of a neighbourhood. Principal neighbourhood aggregation (PNA) combines several statistics and scales them by node degree, retaining information that any single one throws away [37]; degree varies here between three and six, so the scaling is not vacuous.

The other two alternatives question the locality of the model rather than its aggregation. Our layers propagate information one hop at a time, so with four blocks a sensor only sees four rings away. If the missing charge were constrained by the global shape of the light distribution, a wider receptive field would help. We tested this in two forms: a graph U-Net, which pools the graph into a coarser representation before expanding it back [38], and a graph transformer (Graphormer) in which every node attends to every other one regardless of adjacency [39]. We also tested metric edge features, anisotropic hexagonal convolution with one weight per direction, per-channel equalisation, heteroscedastic prediction of mean and variance, and averaging ensembles [40].

Table 1: Training configuration of the reference model.

Parameter	Value
Architecture	HexGNN, 4 blocks, hidden width 48
Trainable parameters	38 305
Activation	GELU
Loss	MSE over the 61 channels
Optimiser	AdamW [42]
Learning rate	10^{-3} , cosine decay [43]
Weight decay	10^{-4}
Batch size	512
Training samples	15 943 000 (single pass)
Validation events	400 000, fixed masks
Model selection	Validation MAE on the imputed channel

2.5 Training

We trained with the mean squared error (MSE) computed over the full 61-channel output rather than the suppressed entries alone,

$$\mathcal{L} = \frac{1}{61} \sum_{c=1}^{61} (\hat{y}_c - y_c)^2, \quad (3)$$

which treats the task as ordinary regression and keeps the healthy channels as a consistency constraint. We selected this loss after comparing it against the mean absolute error (MAE) and the Huber loss [41] with $\delta = 0.1$. We also tested two physics-informed variants, one penalising the displacement of the reconstructed position and one penalising the shift of the energy spectrum; both are reported in Section 3.

Table 1 lists the configuration. Two aspects of the procedure are worth making explicit. First, the training set does not fit in memory, so modules are loaded one at a time and contribute 107 000 events each; the model therefore performs a single pass over 15 943 000 masked samples. Since the masks are regenerated for every sample, no exact input is ever repeated, which is why we applied no early stopping. Second, we selected the final weights by the validation error on the imputed channel, computed on a fixed set of 400 000 events with a fixed mask seed. We did not use the total loss for this purpose. It is dominated by the 60 healthy channels and stays nearly flat across the run, so it separates checkpoints poorly.

Each run stores its preprocessing configuration inside the checkpoint. This is a safeguard rather than a detail: a model trained with one normalisation and evaluated with another produces meaningless numbers without raising any error, and verifying the match automatically removes that failure mode.

2.6 Evaluation Metrics

The quantity the model minimises is the error in the recovered charge, but on its own it says little about whether the method is useful. What matters is how much of the degradation caused by the fault is undone. We therefore evaluated on three levels: the charge itself, the position it produces, and the physical quantities the detector is built to deliver.

At the charge level we report the MAE on the imputed channel in normalised units, together with the bias, which acts as a guardrail: a model can attain a low absolute error while systematically under-predicting, and the bias exposes

that. Because the normalised error is hard to interpret on its own, we also express it relative to the mean charge of each channel. This last figure is sensitive to the number of events used to compute the denominator, so we evaluated numerator and denominator on the same sample throughout.

The position level is our primary metric, since it is the one that reaches the image. For each event we computed the interaction position by CoG weighted by the squared charge [19], [20], in three versions of the same event: intact, with the channel suppressed, and with the channel imputed. Comparing them against the intact position gives two displacements. The *degraded* displacement ΔR_{deg} is how far the centroid moves when the fault goes uncorrected, and therefore measures the damage. The *imputed* displacement ΔR_{imp} is how far it still lies from the true position after the model has acted, and therefore measures the residual error. Recovery is the fraction of the damage the model removes,

$$\text{Recovery} = 100 \times \left(1 - \frac{\Delta R_{\text{imp}}}{\Delta R_{\text{deg}}} \right), \quad (4)$$

with both terms evaluated at the same quantile and over the same population of events. A value of 100 % would mean the imputed position coincides with the intact one, and 0 % that imputing changed nothing. We report the value at the 90th percentile, which weights the difficult events, and also the mean. Averaging over the 61 channels gives the macro figures used throughout, and the minimum over channels characterises the worst case. Since a ratio says nothing about the magnitude of what remains, we also report ΔR_{imp} in millimetres.

Two further analyses ask a different question. Recovery tells us how close the imputed position is to the true one, but not whether the restored signal is physically admissible. A model could place events correctly on average while distorting the quantities the detector is built to measure.

The first analysis concerns spatial resolution. Each of the 364 etched structures produces a spot in the flood map, and the width of that spot sets how finely two interactions can be told apart [9]. We therefore fitted the spot and measured its FWHM in three conditions — intact, degraded and imputed — so that the improvement can be read in the same units in which detector resolution is normally quoted.

The second concerns energy. Suppressing a channel removes part of the collected light, so the reconstructed energy of the event falls and the photopeak shifts towards lower values; a good imputation should put it back. Measuring this is complicated by the absence of energy calibration, since we cannot convert ADC counts into keV. We avoided the problem with a paired differential design: over the same events we subtract the true charge of the channel and add back the prediction, so any calibration factor multiplies both terms and cancels. We evaluated this by zones of the crystal, defined as concentric hexagonal bands, because the absorbing adhesive on the lateral faces makes the periphery behave differently from the centre [21] and an average over the whole crystal would hide that difference.

Table 2: Architecture comparison. Common campaign of 600 000 events per test file. Recovery in per cent, MAE in normalised units.

Model	Params	Rec. p90	Rec. mean	MAE
DeepMLP	500 541	51.39	45.90	0.7219
ResidualMLP	345 981	48.92	44.81	0.7299
Graphormer	4 263 881	55.17	49.65	0.6600
Graph U-Net	331 649	55.22	47.95	0.6918
PNA	165 761	58.45	52.48	0.6309
HEXGNN-S	38 305	58.02	52.66	0.6195
HEXGNN-M	225 217	58.27	52.64	0.6185
HEXGNN-L	398 593	57.47	51.96	0.6385

Finally, a note on what these numbers can and cannot support. Comparisons between models are paired, in that every model is evaluated on the same modules, the same events and the same masks, so relative differences are measured precisely. Absolute performance on a new detector, however, is a different matter, because modules differ from one another far more than training runs do. We quantified both sources of variation separately, and Section 3 reports them alongside the results they qualify.

3 RESULTS

3.1 Value of the Geometric Prior

Table 2 compares the architectures under a common evaluation campaign of 600 000 events per test file. The two models without a geometric prior reach 51.39 % and 48.92 % recovery at the 90th percentile, while the smallest graph model reaches 58.02 % with an order of magnitude fewer parameters. Widening the graph model, however, changes little: 58.27 % for HEXGNN-M and 57.47 % for HEXGNN-L, against 58.02 % for the 38 k-parameter reference.

To test whether the graph structure itself carries the advantage, rather than the inductive bias of message passing in general, we retrained the reference with the adjacency randomly permuted, preserving the number of nodes, the number of edges and the degree distribution. Recovery falls to 28.08 %, 27.66 % and 31.28 % across three shuffles (Fig. 5), against 58.02 % for the true adjacency.

Against the classical methods, evaluated on the same channels, the network recovers 57.9 % compared with 45.41 % for the neighbour mean and 46.56 % for the per-channel linear regression, differences of 12.49 and 11.33 percentage points respectively.

We also quantified how far the useful information extends. Fitting linear regressions restricted to concentric rings of neighbours gives 44.04 % recovery with the first ring (5.1 channels on average), 46.42 % with the second (13.7 channels) and 46.51 % with the third (24.1 channels), against 46.55 % when all 60 remaining channels are used.

3.2 Reference Performance and Its Variability

A single figure of merit says little without knowing how much it moves, so we measured two sources of variability separately. Table 3 reports both. Across four independent

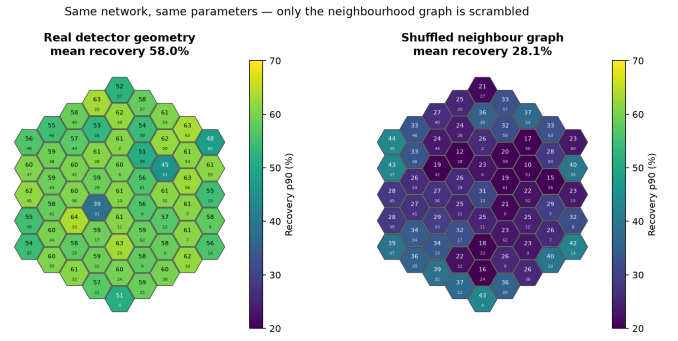


Figure 5: Effect of destroying the detector geometry. Recovery at the 90th percentile for the reference model with the true adjacency and for three independent random permutations of the graph that preserve node count, edge count and degree distribution.

Table 3: Reference model and sources of variability. Values are mean $\pm$ standard deviation. n is the number of independent runs.

Source	n	Rec. p90	Rec. mean
Independent training runs	4	58.04 ± 0.37	52.15 ± 0.31
Data partition (CV)	5	56.56 ± 2.32	50.60 ± 2.04

training runs, differing in the global seed and in the module ordering, recovery at the 90th percentile is 58.04 ± 0.37 % and the mean recovery is 52.15 ± 0.31 %.

However, repeating the split with five different partitions of the same 159 modules gives 56.56 ± 2.32 %, a standard deviation six times larger than the one observed between training runs of a fixed partition. Measuring the model on individual unseen modules makes the origin of that spread explicit: recovery varies between 29.2 % and 63.0 % depending on the module, with a standard deviation of 7.86 percentage points (Fig. 6), which places the standard error of a mean over five test modules at 3.5 points. The dominant uncertainty in this work is therefore which detectors happen to fall in the test partition, not which weights the optimiser converged to.

Performance is not uniform across the detector. Fig. 7 maps the recovery of each of the 61 channels. The worst channel of the reference band averages 42.29 ± 4.45 % over the four runs, with individual values spanning 37.54 % to 47.76 %; the channel attaining that minimum is not the same in every run, alternating between Ich 60 and Ich 31.

The effect of imputation on the reconstructed image is shown in Fig. 8, which compares the flood map of a test module with one channel suppressed against the same map after imputation.

3.3 Multiple Simultaneous Failures

Training with a single suppressed channel does not transfer to multiple failures. Table 4 reports recovery averaged over $k = 1 \dots 4$ suppressed channels for the three spatial regimes. A model trained with $k = 1$ falls to 33.74 % on contiguous

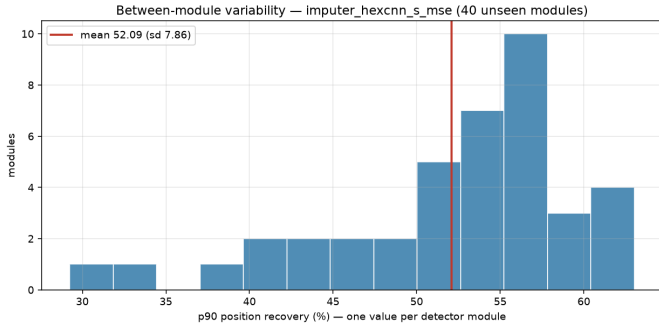


Figure 6: Recovery at the 90th percentile measured on individual unseen training modules. The spread between detectors (standard deviation of 7.86 percentage points) is an order of magnitude larger than the spread between training runs on a fixed partition.

Table 4: Recovery at the 90th percentile by failure regime, averaged over $k = 1 \dots 4$ suppressed channels. Campaign of 20 seeds and 200 000 events per file.

Training regime	Cluster	Near	Scatter
Single channel ($n=3$)	33.74 ± 11.68	38.28 ± 9.23	47.69 ± 4.95
Cluster, $k=1-4$ ($n=1$)	58.46	52.62	52.44
Near, $k=1-4$ ($n=3$)	58.31 ± 0.51	54.61 ± 0.58	53.43 ± 0.67

failures, and its standard deviation across runs rises to 11.68 points, since it performs adequately at $k = 1$ and collapses at $k = 4$. Models trained with k drawn from 1 to 4 stay between 53 % and 58 % in all three regimes, with standard deviations below 0.7 points.

Between the two multi-channel regimes, however, the differences are small. The model trained on the *near* regime exceeds the one trained on contiguous clusters by 1.99 points in its own regime, a difference of 2.9 standard deviations, while the two are indistinguishable on contiguous failures (-0.14 points) and on scattered ones ($+0.99$ points, 1.3 standard deviations). Fig. 9 shows the three regimes side by side and Fig. 10 the degradation as k grows.

3.4 Physical Fidelity

Beyond the position error, we measured whether imputation preserves the two physical quantities the detector is built to deliver.

The point-spread function of the flood map narrows from a macro-averaged FWHM of 0.1656 mm with a suppressed channel to 0.0847 mm after imputation, a recovery of 48.82 % of the induced blur (Fig. 11). The positional bias falls from 0.1262 mm to 0.0218 mm, a recovery of 82.71 %.

The energy spectrum is preserved to a comparable degree and with little dependence on the region of the crystal: spectral recovery is 83.25 % in the core, 84.32 % in the middle band and 81.19 % at the edge, for a global value of 82.92 % (Fig. 12). The residual shift of the photopeak stays between 0.20 and 0.24 ADC units across the three zones.

3.5 Physics-Informed Losses and Ensembles

Adding a term that penalises the displacement of the reconstructed position leaves recovery unchanged at 58.23 % against 58.02 % for the reference. Adding a term on the energy spectrum degrades it to 46.04 %, with the bias changing sign. Heteroscedastic prediction of mean and variance reaches 20.08 %.

Averaging the outputs of three independently trained replicas raises recovery to 58.69 %, and averaging three different architectures raises it to 58.73 %, against 58.04 ± 0.37 % for a single model.

4 DISCUSSION

Interpretacion, comparacion con el estado del arte y limitaciones.

5 CONCLUSIONS

Conclusiones y lineas futuras.

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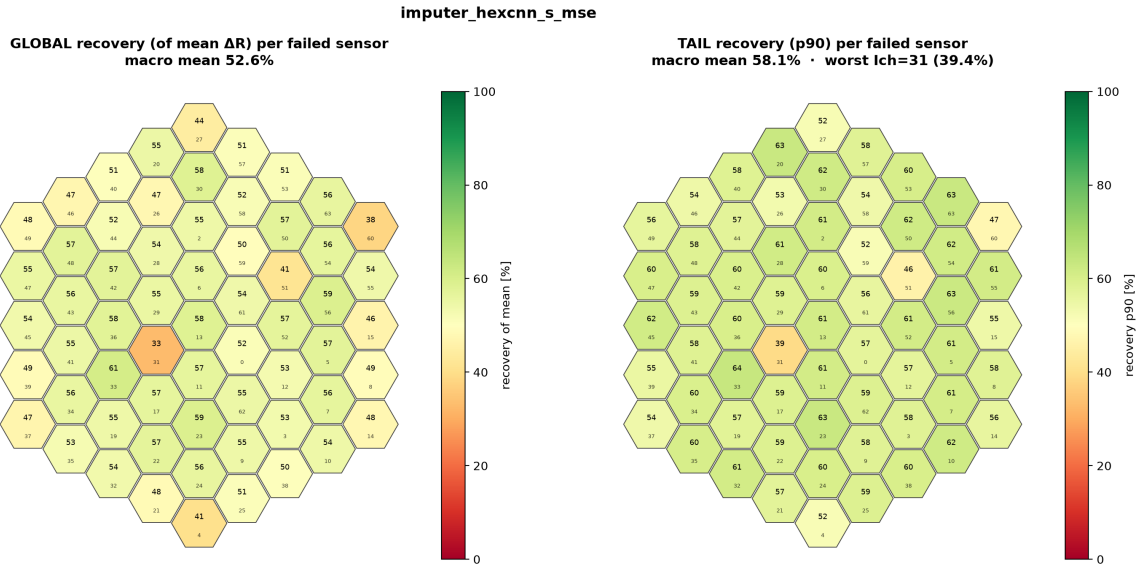


Figure 7: Per-channel recovery over the 61 active sensors. Left: recovery at the 90th percentile. Right: mean recovery. Channels at the perimeter of the hexagonal array recover less charge than central ones.

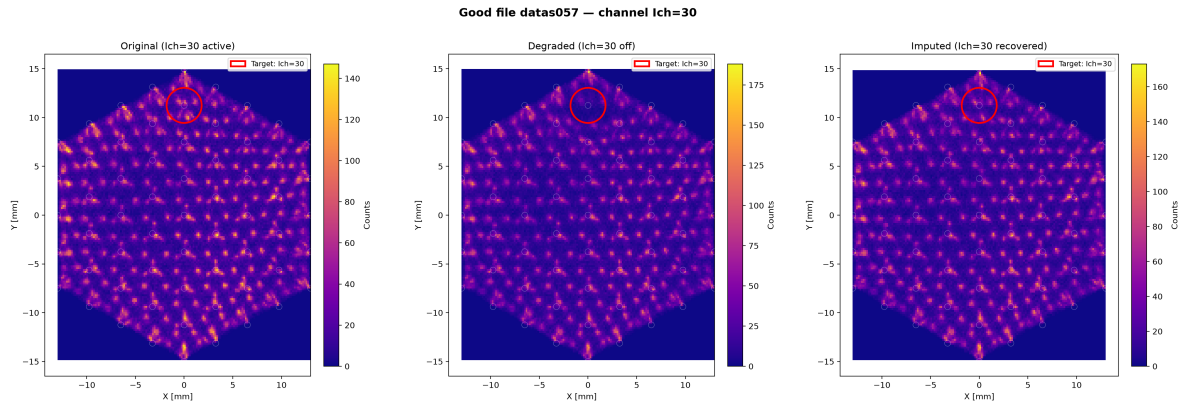


Figure 8: Flood map of a test module. From left to right: reference map with all channels active, map with one channel suppressed, and map recovered by the model.



Figure 9: Recovery by failure regime for models trained under each of the three mask-generation strategies.

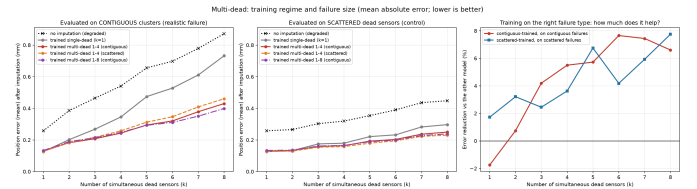


Figure 10: Recovery as a function of the number of simultaneously suppressed channels, for models trained with a single channel and with one to four.

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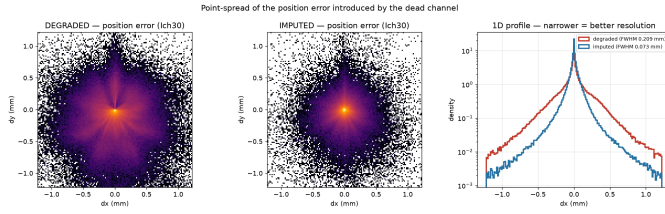


Figure 11: Point-spread function of the flood map with all channels active, with one channel suppressed, and after imputation.

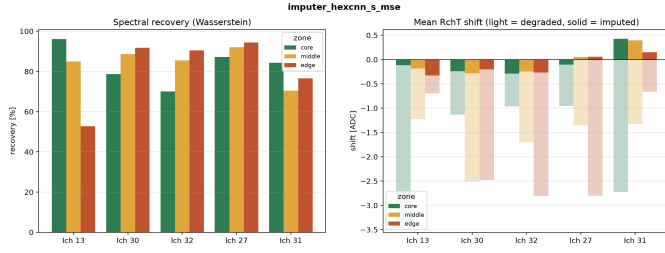


Figure 12: Energy spectrum by crystal zone, comparing the original distribution with the degraded and imputed ones. Zones are defined by concentric hexagonal bands.

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